ENTERPRISE AI DATA ANALYST

Deep Learning-Based Business Analytics Using Transformer and Natural Language Processing

Project Report

Submitted in Partial Fulfillment of the Requirements for the Award of the Diploma In

Artificial Intelligence and Data Science

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Academic Year: 2025–2026

Abstract

Enterprise AI Data Analyst is a deep learning–based business analytics application developed to help users analyze business datasets using natural language queries. Traditional business intelligence tools often require users to have knowledge of SQL, data visualization software, or programming languages. This project addresses that challenge by allowing users to interact with business data through simple English questions.

The system is developed using **Python** and **Streamlit** for the user interface, while **PyTorch** is used to implement a Transformer-based intent classification model. The application accepts CSV and Excel datasets, automatically preprocesses and normalizes the data, predicts the user's intent, performs the requested business analysis, generates interactive visualizations using Plotly, and provides AI-generated business insights. It also supports exporting analytical results in PDF, CSV, and Excel formats.

The system demonstrates how Natural Language Processing (NLP), Deep Learning, and Business Intelligence can be integrated into a single application to simplify business data analysis. Although the current implementation supports a predefined set of business analytics queries, its modular architecture allows additional intents and analytics modules to be incorporated in future versions.

The project successfully provides an intelligent, interactive, and user-friendly platform for business analytics, reducing the technical expertise required to obtain meaningful insights from business data.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's digital era, organizations generate vast amounts of business data through daily operations such as sales, customer transactions, inventory management, and financial activities. Extracting meaningful insights from this data is essential for improving business performance, supporting strategic decision-making, and gaining a competitive advantage. However, traditional business intelligence tools often require technical expertise in database management, programming languages, or data visualization software, making them less accessible to non-technical users.

Artificial Intelligence (AI) and Natural Language Processing (NLP) have transformed the way users interact with computer systems. Modern deep learning techniques, especially Transformer-based models, enable computers to understand natural language with high accuracy. This advancement allows users to retrieve business information simply by asking questions in plain English instead of writing complex queries or manually analyzing reports.

The **Enterprise AI Data Analyst** is developed to simplify business data analysis by combining Artificial Intelligence, Deep Learning, and Business Intelligence techniques into a single application. The system enables users to upload business datasets in CSV or Excel format, ask analytical questions using natural language, and receive meaningful insights automatically. A Transformer-based intent classification model identifies the user's request, while the analytics engine processes the dataset and generates relevant results. Interactive visualizations, business insights, and report export features further enhance the user experience.

The application is implemented using **Python** as the programming language, **PyTorch** for the deep learning model, **Streamlit** for the web-based user interface, **Pandas** for data processing, and **Plotly** for interactive visualizations. The system also supports exporting analysis results in PDF, CSV, and Excel formats, making it suitable for business reporting and decision support.

This project demonstrates how Artificial Intelligence can simplify business analytics by enabling users to interact with data using natural language while producing accurate analytical results through an intuitive and user-friendly interface.

1.2 Problem Statement

Many organizations collect large amounts of business data but face difficulties in analyzing it efficiently. Existing business intelligence tools often require technical knowledge of SQL, programming, or specialized visualization software, limiting their usability for non-technical users. As a result, generating business reports and obtaining insights can become time-consuming and dependent on technical personnel.

Furthermore, most traditional analytics systems require users to manually select analysis methods and visualization techniques before meaningful insights can be obtained. This reduces productivity and increases the complexity of data analysis.

Therefore, there is a need for an intelligent business analytics system that allows users to upload datasets, ask questions in natural language, automatically understand the user's intent, perform the required business analysis, generate appropriate visualizations, and provide meaningful business insights through an easy-to-use interface.

1.3 Objectives

The main objective of this project is to develop an intelligent business analytics application that enables users to analyze business datasets using natural language queries.

The specific objectives are:

- To develop a Transformer-based intent classification model for understanding business queries.
- To design an interactive web application for business data analysis.
- To automatically preprocess and normalize uploaded datasets.
- To generate business analytics based on user queries.
- To provide interactive data visualizations for analytical results.
- To generate AI-based business insights.
- To support exporting analytical reports in PDF, CSV, and Excel formats.

- To provide a user-friendly platform suitable for both technical and non-technical users.

1.4 Scope of the Project

The scope of this project is focused on developing an AI-powered business analytics application capable of processing structured business datasets provided in CSV and Excel formats. The system supports natural language interaction by identifying user intent through a Transformer-based deep learning model and performing predefined business analytics.

The application provides data preprocessing, KPI dashboards, interactive visualizations, business insights, and report export functionality. The current implementation supports a predefined set of business analysis queries and is designed with a modular architecture, allowing additional analytical functions and business intents to be incorporated in future versions.

1.5 Report Organization

This report is organized into six chapters. Chapter 1 introduces the project, including the problem statement, objectives, and scope. Chapter 2 presents the literature review related to business intelligence, natural language processing, and Transformer models. Chapter 3 describes the system analysis and design, including the proposed architecture. Chapter 4 explains the methodology, implementation, and technologies used to develop the system. Chapter 5 discusses the experimental results, system outputs, and performance evaluation. Finally, Chapter 6 presents the conclusion and future enhancements of the project.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

Business Intelligence (BI), Artificial Intelligence (AI), and Natural Language Processing (NLP) have significantly transformed the way organizations analyze and interpret business data. Traditional BI systems primarily depend on structured queries, predefined reports, and manual dashboard creation, which often require technical expertise. Recent developments in AI and deep learning have enabled more intelligent and user-friendly analytics systems capable of understanding natural language and generating meaningful insights automatically.

This chapter reviews the concepts, technologies, and previous research related to business analytics, natural language processing, Transformer models, and interactive data visualization that form the foundation of the proposed **Enterprise AI Data Analyst** system.

2.2 Business Intelligence

Business Intelligence refers to the technologies and processes used to collect, analyze, and visualize business data to support organizational decision-making. BI systems enable organizations to monitor key performance indicators (KPIs), identify trends, evaluate business performance, and generate reports.

Traditional BI tools such as Microsoft Power BI, Tableau, and Qlik Sense provide powerful visualization capabilities. However, these platforms often require users to manually create dashboards or possess knowledge of data modeling and visualization techniques. This creates a challenge for non-technical users who need quick access to business insights.

The proposed system addresses this limitation by allowing users to analyze business data through natural language questions rather than manually designing reports.

2.3 Natural Language Processing

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language. NLP combines computational linguistics with machine learning to process textual information.

In business analytics, NLP allows users to communicate with analytical systems using everyday language instead of complex database queries.

Typical NLP tasks include:

- Text preprocessing
- Tokenization
- Intent classification
- Named Entity Recognition (NER)
- Text classification
- Question answering

In this project, NLP techniques are used to clean user queries, tokenize text, classify business intents, and route requests to the appropriate analytics functions.

2.4 Deep Learning and Transformer Models

Deep Learning has become one of the most successful approaches for solving complex Natural Language Processing problems. Among various deep learning architectures, the **Transformer** model has achieved remarkable performance in language understanding tasks.

Unlike traditional Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, Transformers process entire sequences simultaneously using a mechanism called **Self-Attention**. This enables the model to capture long-range relationships between words while improving training efficiency.

The Transformer model offers several advantages:

- Better understanding of contextual information.
- High accuracy in intent classification.
- Faster parallel processing.
- Improved scalability for larger datasets.

The Enterprise AI Data Analyst uses a Transformer-based intent classifier developed using **PyTorch** to identify user intentions from natural language business queries.

2.5 Data Visualization

Data visualization converts analytical results into graphical representations that improve understanding and support decision-making.

Interactive visualizations help users identify trends, patterns, and anomalies more effectively than numerical reports alone.

The project uses **Plotly Express** to generate interactive charts, including:

- Line Charts
- Bar Charts
- KPI Cards

These visualizations enable users to interpret analytical results quickly and improve overall usability.

2.6 Streamlit for Interactive Applications

Streamlit is an open-source Python framework used to build interactive web applications for data science and machine learning projects.

It offers several advantages:

- Rapid application development
- Interactive widgets
- Built-in data visualization support
- Easy deployment
- Integration with machine learning models

In this project, Streamlit provides the graphical user interface for uploading datasets, entering natural language queries, displaying analytical results, and exporting reports.

2.7 Related Work

Several researchers have proposed intelligent systems that integrate Artificial Intelligence with Business Intelligence to simplify data analysis.

Recent studies have demonstrated the effectiveness of Transformer-based models in improving intent classification and natural language understanding. Similarly, modern business analytics platforms increasingly incorporate AI techniques to automate reporting and provide intelligent recommendations.

However, many existing solutions either focus solely on visualization or require advanced technical knowledge for querying datasets. The proposed Enterprise AI Data Analyst combines deep learning, business analytics, natural language processing, interactive visualization, and report generation into a single integrated platform, making business data analysis more accessible to users with varying technical backgrounds.

2.8 Chapter Summary

This chapter reviewed the fundamental concepts related to Business Intelligence, Natural Language Processing, Deep Learning, Transformer models, Streamlit, and interactive data visualization. It also discussed previous research and existing technologies relevant to the proposed system. These concepts provide the theoretical foundation for the design and implementation of the Enterprise AI Data Analyst presented in the following chapters.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

System analysis and design is an essential phase in software development that focuses on identifying user requirements and designing a system capable of fulfilling those requirements efficiently. This chapter presents the analysis of the existing business analytics process, identifies its limitations, and describes the proposed Enterprise AI Data Analyst system. It also explains the overall system architecture, functional requirements, non-functional requirements, and workflow of the proposed solution.

3.2 Existing System

Traditional business analytics systems mainly rely on Business Intelligence (BI) tools such as Microsoft Excel, Power BI, Tableau, and SQL-based reporting systems. Although these tools provide powerful analytical capabilities, they often require users to possess technical knowledge in database querying, dashboard development, or programming.

Limitations of the Existing System

- Requires technical knowledge to perform data analysis.
- Users must manually create reports and visualizations.
- Limited natural language interaction.
- Time-consuming report generation.
- Business users depend on technical experts for complex analysis.

3.3 Proposed System

The proposed **Enterprise AI Data Analyst** is an intelligent business analytics application that enables users to upload business datasets and analyze them using natural language queries.

The system automatically:

- Uploads and preprocesses datasets.
- Identifies business-related columns.
- Predicts user intent using a Transformer model.
- Performs business analytics.
- Generates interactive charts.
- Produces AI-based business insights.
- Exports reports in PDF, CSV, and Excel formats.

This approach simplifies business analytics for both technical and non-technical users.

3.4 Functional Requirements

The system shall:

- Upload CSV and Excel datasets.
- Normalize dataset columns automatically.
- Accept business questions in natural language.
- Predict user intent using a Transformer model.
- Perform business analytics.
- Display KPI dashboards.
- Generate interactive visualizations.
- Provide AI-generated business insights.
- Export results in PDF, CSV, and Excel formats.

3.5 Non-Functional Requirements

Performance

The application should generate results within a few seconds for normal-sized business datasets.

Usability

The interface should be simple and user-friendly so that non-technical users can operate the system.

Reliability

The system should process valid datasets consistently and produce accurate analytical results.

Scalability

The modular architecture allows additional business intents and analytics modules to be incorporated in future versions.

3.6 System Architecture

The overall architecture of the proposed system consists of the following modules:

User

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Streamlit Web Interface

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Dataset Upload Module

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Dataset Preprocessing

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NLP Query Preprocessing

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Transformer Intent Classification

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Analytics Engine

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KPI Dashboard Charts AI Insights

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Export Reports

3.7 System Workflow

The workflow of the proposed system is as follows:

1. The user uploads a CSV or Excel dataset.
2. The dataset is preprocessed and normalized.
3. The user enters a business question.
4. The NLP module preprocesses the query.
5. The Transformer model predicts the user's intent.
6. The analytics engine performs the requested analysis.
7. The visualization module generates appropriate charts.
8. AI business insights are generated.
9. Results can be exported in PDF, CSV, or Excel format.

3.8 Chapter Summary

This chapter discussed the analysis and design of the proposed Enterprise AI Data Analyst system. It presented the limitations of existing systems, the proposed solution, functional and non-functional requirements, system architecture, and workflow. The next chapter explains the methodology and implementation used to develop the application.

CHAPTER 4

METHODOLOGY AND IMPLEMENTATION

4.1 Introduction

This chapter explains the methodology adopted for developing the **Enterprise AI Data Analyst** system and describes the implementation of its major modules. The system integrates Artificial Intelligence (AI), Natural Language Processing (NLP), Deep Learning, Business Analytics, and Data Visualization into a single platform that enables users to analyze business datasets through natural language queries.

4.2 Development Methodology

The project follows an **Iterative Development Methodology**, where the application was developed, tested, and improved in multiple stages. Each module was implemented independently and then integrated into the complete system.

The major development phases are:

- Requirement Analysis
- Dataset Collection
- Data Preprocessing
- Transformer Model Development
- Analytics Engine Development
- Visualization Module Development
- Report Generation
- System Testing and Validation

This methodology helped identify and fix issues during development while allowing new features to be integrated gradually.

4.3 System Modules

The proposed system consists of the following modules.

4.3.1 Dataset Upload Module

The application allows users to upload business datasets in **CSV** and **Excel** formats.

Functions

- Upload CSV datasets
- Upload Excel datasets
- Read datasets using Pandas
- Store datasets for further analysis

4.3.2 Data Preprocessing Module

Before analysis, the uploaded dataset undergoes preprocessing to ensure consistency.

The preprocessing process includes:

- Removing unnecessary spaces
- Handling missing values
- Standardizing column names
- Dataset normalization
- Automatic business column mapping

This enables the system to work with datasets having different column names.

4.3.3 Natural Language Processing Module

The NLP module processes the user's business question before prediction.

The preprocessing steps include:

- Text cleaning
- Tokenization

- Vocabulary lookup
- Sequence padding

The processed query is then passed to the Transformer model.

4.3.4 Transformer-Based Intent Classification

A Transformer model built using **PyTorch** predicts the user's business intent.

Example:

Input Query

Show monthly sales

Predicted Intent

Monthly_sales

The predicted intent is forwarded to the Analytics Engine for processing.

4.3.5 Analytics Engine

The Analytics Engine performs business calculations based on the predicted intent.

The implemented analytics include:

- Total Sales
- Average Sales
- Total Profit
- Average Profit
- Top Customers
- Top Products
- Monthly Sales
- Yearly Sales
- Dashboard Summary

The results are generated dynamically from the uploaded dataset.

4.3.6 Visualization Module

The Visualization module automatically converts analytical results into interactive charts using **Plotly**.

The implemented chart types include:

- Line Charts
- Bar Charts
- KPI Cards

The visualization is selected automatically based on the analysis result.

4.3.7 AI Business Insights Module

The AI Insights module generates short business observations from the analytical results.

Examples include:

- Sales performance summaries
- Customer performance observations
- Product performance analysis
- Business recommendations

This helps users interpret the results more effectively.

4.3.8 Report Export Module

The system supports exporting analytical reports in multiple formats.

Supported formats:

- PDF
- CSV
- Excel

This allows users to save and share analytical reports easily.

4.4 Technologies Used

Technology	Purpose
Python	Core programming language
PyTorch	Transformer model implementation
Streamlit	Web application interface
Pandas	Data preprocessing and analysis
Plotly	Interactive data visualization
ReportLab	PDF report generation
OpenPyXL	Excel file handling

4.5 System Workflow

The overall workflow of the proposed system is shown below.

User

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Upload Dataset

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Data Preprocessing

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```
Natural Language Query
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   ▼
Text Preprocessing
   |
   ▼
Transformer Intent Prediction
   |
   ▼
Analytics Engine
   |
   ├── KPI Dashboard
   ├── Charts
   ├── AI Insights
   └── Export Reports
   |
   ▼
Display Results
```

4.6 Implementation

The application is implemented as a collection of modular Python components.

The major modules include:

- **Transformer Module** – Predicts user intent.
- **Analytics Module** – Performs business calculations.

- **Visualization Module** – Generates interactive charts.
- **Dashboard Module** – Displays KPI metrics.
- **Export Module** – Creates PDF, CSV, and Excel reports.
- **Streamlit Interface** – Provides an interactive web application for users.

This modular design improves maintainability and allows new analytical features to be added with minimal changes.

4.7 Advantages of the Proposed System

The proposed system offers several advantages:

- Supports natural language business queries.
- Uses a Transformer-based deep learning model for intent prediction.
- Automatically preprocesses uploaded datasets.
- Generates interactive visualizations.
- Provides AI-generated business insights.
- Supports PDF, CSV, and Excel export.
- Offers a simple interface suitable for non-technical users.

4.8 Chapter Summary

This chapter presented the methodology and implementation of the **Enterprise AI Data Analyst** system. It described the development methodology, major software modules, technologies used, system workflow, and implementation details. The next chapter will present the **Results and Discussion**, including screenshots of the application, output analysis, and system evaluation.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results obtained from the implementation of the **Enterprise AI Data Analyst** system. It demonstrates the functionality of the developed application through various test cases, screenshots, and output analysis. The chapter also discusses the system's performance, evaluates whether the project objectives were achieved, and highlights the effectiveness of the proposed solution in analyzing business datasets using natural language queries.

5.2 Experimental Setup

The application was developed and tested using the following software and hardware environment.

Software Requirements

Software	**Version**
Operating System	Windows 11
Python	3.x
Streamlit	Latest Stable Version
PyTorch	Latest Stable Version
Pandas	Latest Stable Version
Plotly	Latest Stable Version
ReportLab	Latest Stable Version

Hardware Requirements

Component	Specification
Processor	Intel Core i5 or higher
RAM	8 GB or higher
Storage	256 GB SSD or higher

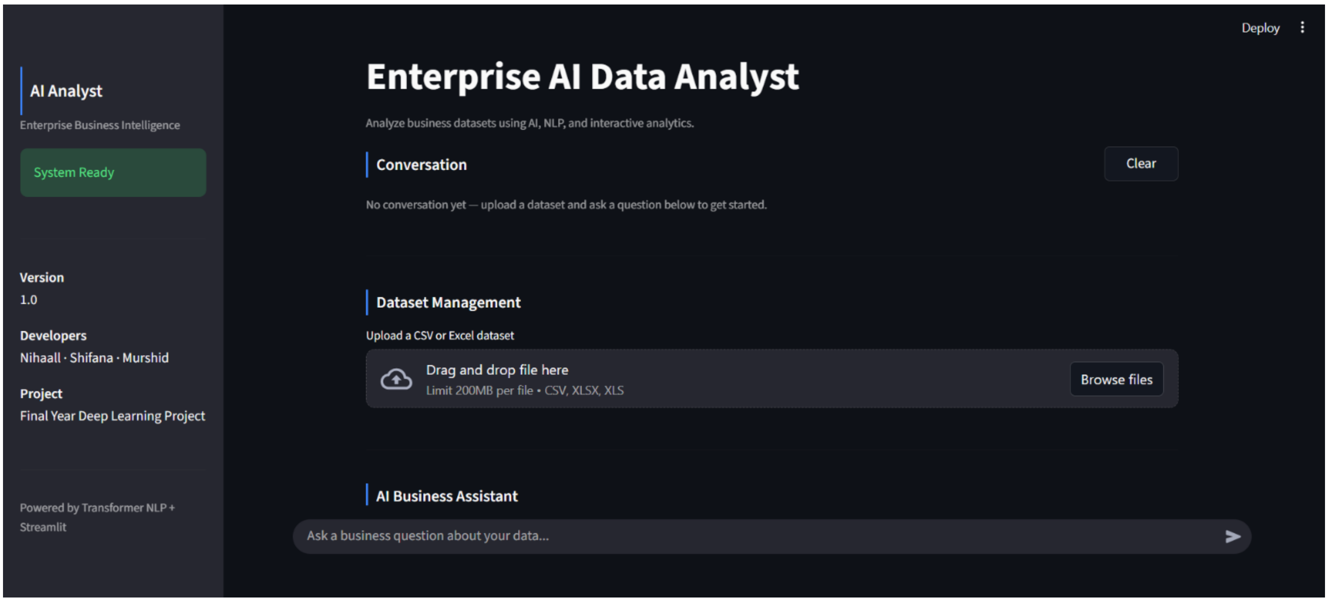
5.3 System Execution

The developed application successfully performs the following operations:

- Uploads CSV and Excel datasets.
- Automatically preprocesses and normalizes datasets.
- Accepts business questions in natural language.
- Predicts user intent using a Transformer-based model.
- Performs business analytics.
- Displays KPI dashboards.
- Generates interactive visualizations.
- Produces AI-generated business insights.
- Exports reports in PDF, CSV, and Excel formats.

5.4 Output Screenshots

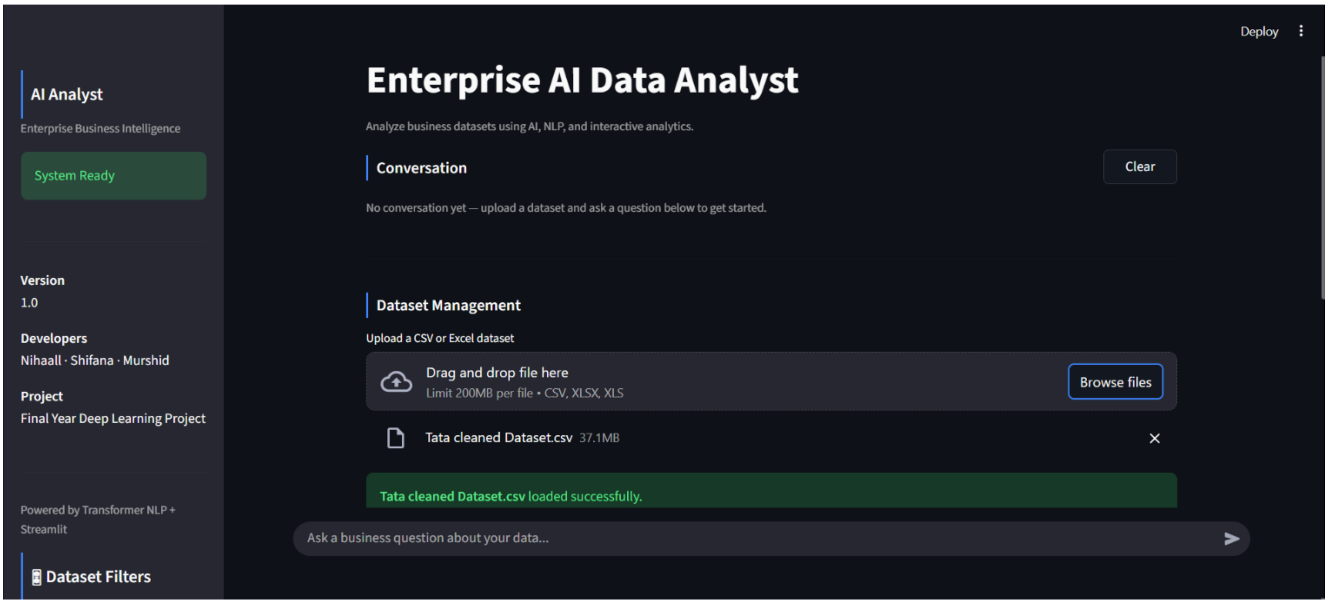
Figure 5.1 Home Page



Description:

The home page provides the user interface for uploading datasets and interacting with the AI-powered business analyst.

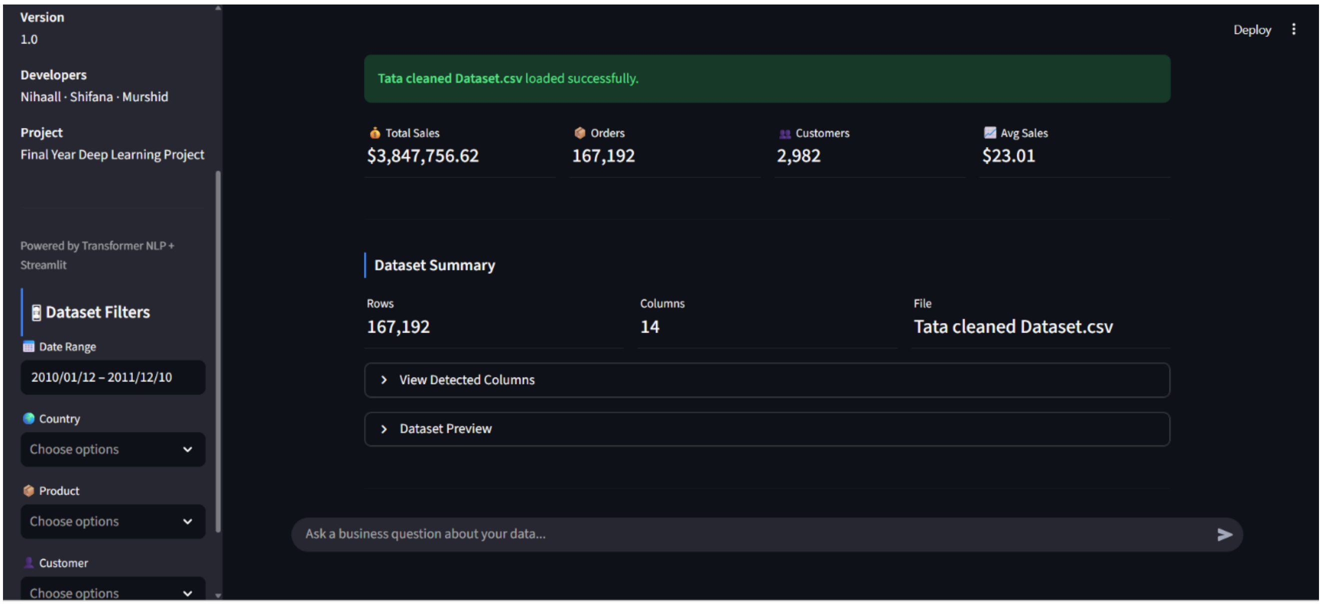
Figure 5.2 Dataset Upload



Description:

The application accepts CSV and Excel files and automatically preprocesses the uploaded dataset for analysis.

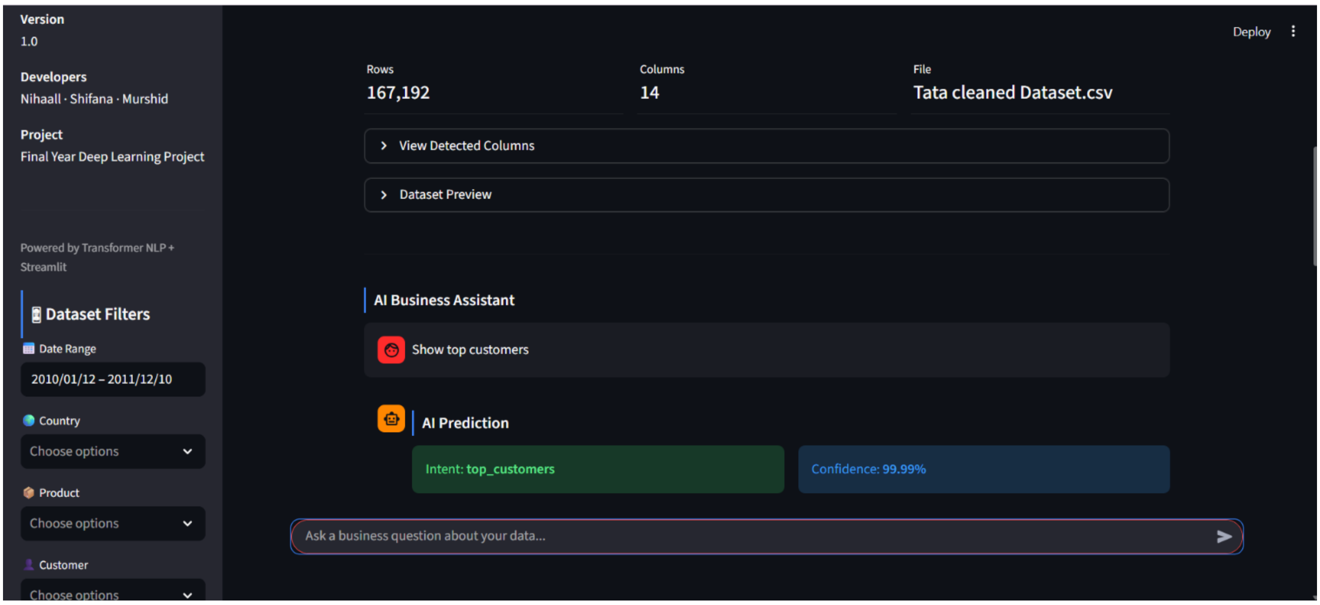
Figure 5.3 KPI Dashboard



Description:

The dashboard displays key business metrics such as Total Sales, Total Orders, Customer Count, and Product Count.

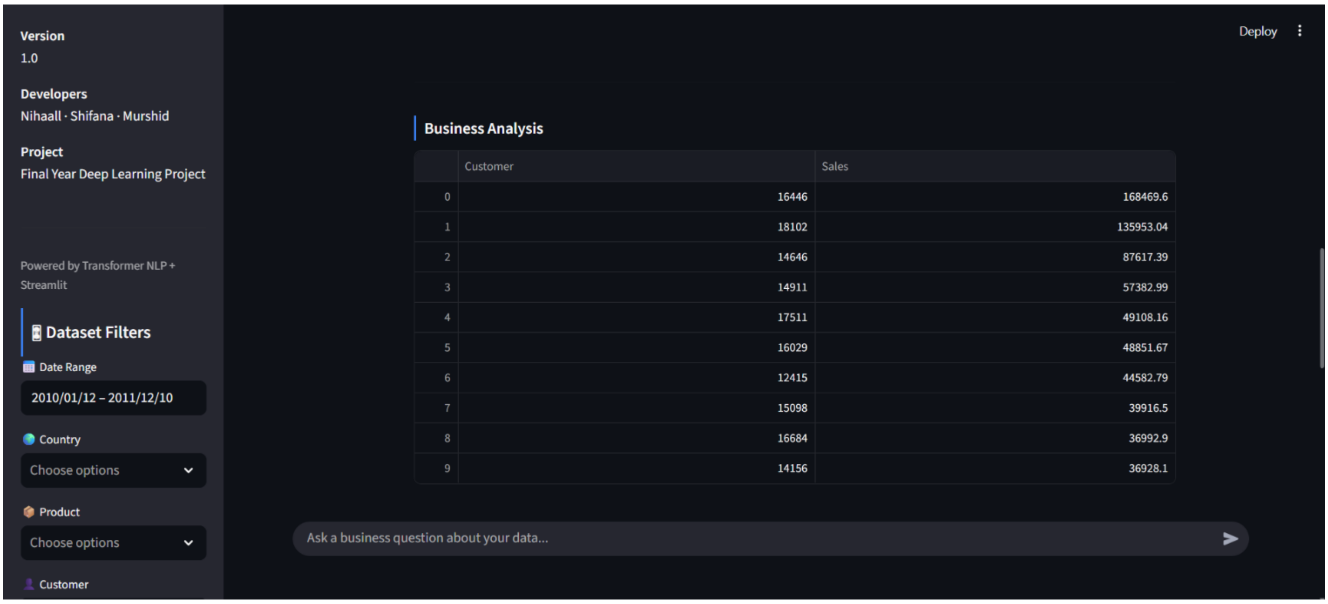
Figure 5.4 Natural Language Query



Description:

The user asks a business question in natural language, and the Transformer model predicts the corresponding business intent.

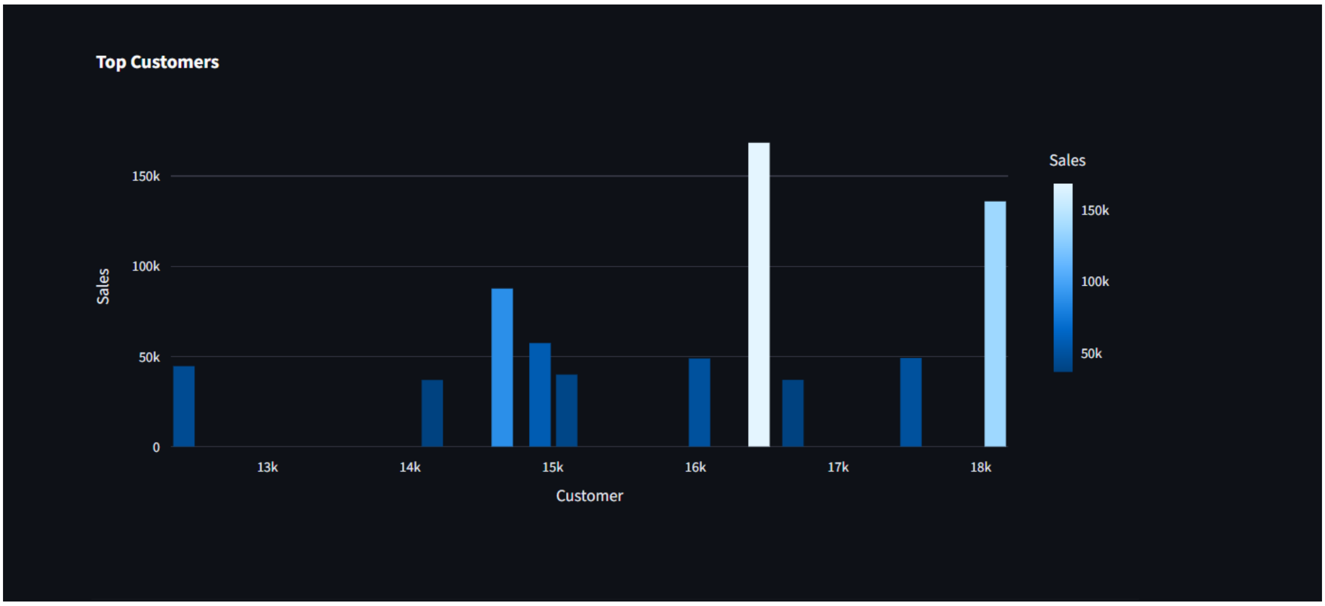
Figure 5.5 Business Analysis Result



Description:

The analytics engine processes the predicted intent and generates business results dynamically from the uploaded dataset.

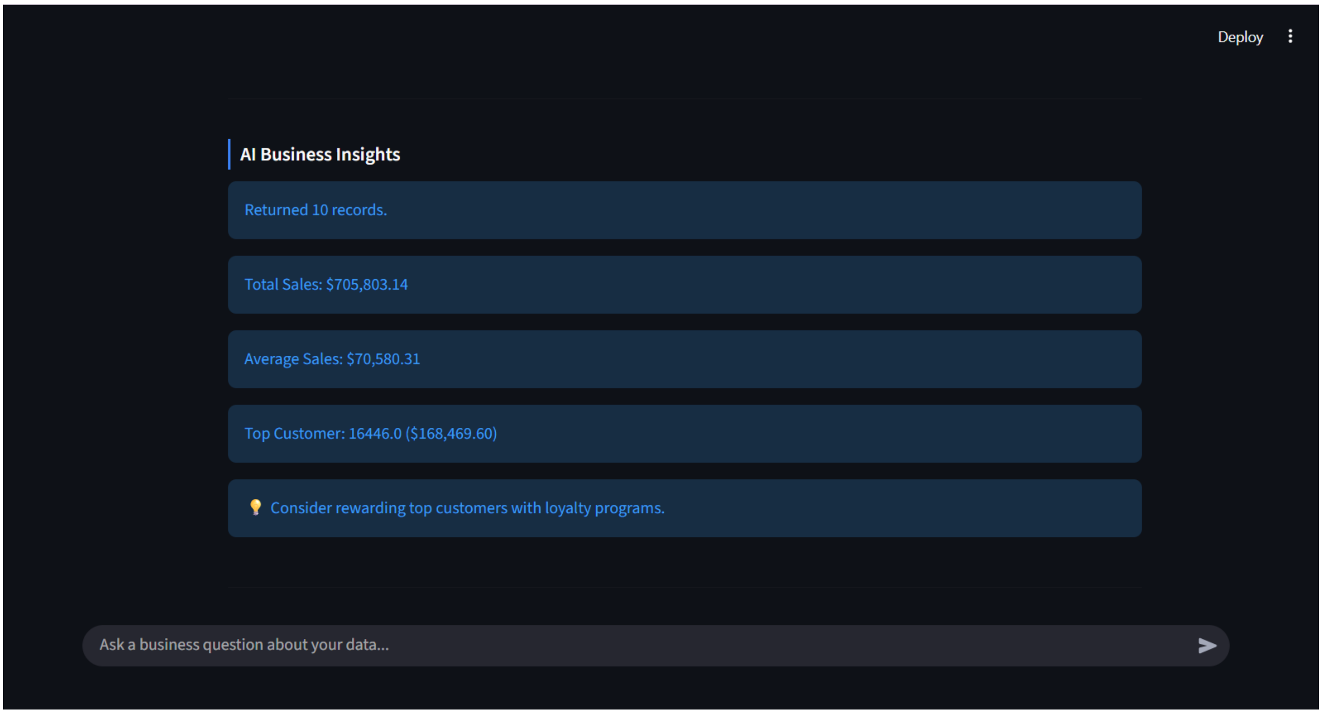
Figure 5.6 Interactive Visualization



Description:

The visualization module automatically generates interactive charts that help users understand business trends and performance.

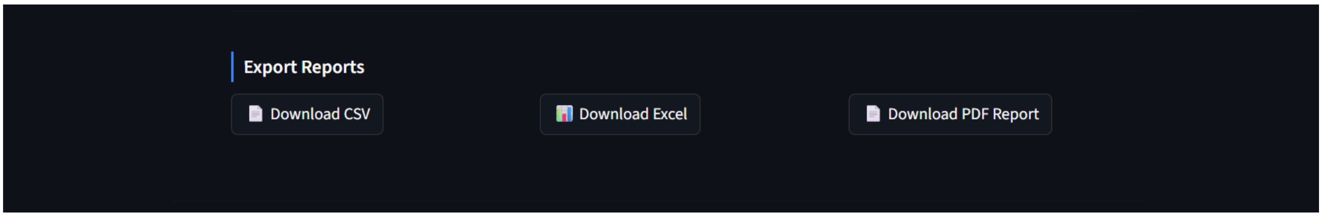
Figure 5.7 AI Business Insights



Description:

The application generates concise business insights to assist users in interpreting analytical results.

Figure 5.8 Export Reports



Description:

Users can export analytical results in PDF, CSV, and Excel formats for reporting and further analysis.

5.5 Discussion

The developed system successfully demonstrates the integration of Deep Learning, Natural Language Processing, Business Analytics, and Data Visualization into a single application. The Transformer-based intent classifier effectively identifies supported business queries, allowing users to interact with datasets using natural language.

The automatic preprocessing module enables the system to work with different business datasets by normalizing column names before analysis. Interactive dashboards and visualizations improve the readability of analytical results, while AI-generated insights help users interpret business performance more effectively.

The export functionality further enhances the usefulness of the application by allowing analytical reports to be saved in multiple formats.

5.6 Limitations

Although the developed system performs effectively for the implemented analytics, several limitations remain:

- The application currently supports a predefined set of business analytics queries.
- Queries outside the implemented intents are not processed automatically.
- The system is designed for structured datasets only.
- Performance depends on the quality and completeness of the uploaded dataset.
- The current implementation is intended for single-user operation.

These limitations provide opportunities for future improvements.

5.7 Chapter Summary

This chapter presented the implementation results and evaluated the performance of the **Enterprise AI Data Analyst** system. The experimental results demonstrate that the application successfully supports natural language-based business analytics, interactive visualizations, AI-generated insights, and report export functionalities. Overall, the project objectives were achieved within the defined scope.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The **Enterprise AI Data Analyst** project successfully demonstrates the integration of **Artificial Intelligence (AI)**, **Natural Language Processing (NLP)**, **Deep Learning**, and **Business Analytics** into a single intelligent application. The developed system enables users to upload structured business datasets and analyze them through natural language queries without requiring advanced technical knowledge.

A Transformer-based intent classification model was implemented to understand user queries and identify the appropriate business intent. The application automatically preprocesses uploaded datasets, performs business analytics, generates interactive visualizations, displays key performance indicators (KPIs), and provides AI-generated business insights. Additionally, the system supports exporting analytical reports in PDF, CSV, and Excel formats, making it useful for reporting and decision-making.

The project achieved its primary objectives of developing an intelligent, user-friendly business analytics platform that simplifies data analysis through natural language interaction. The modular architecture adopted during development also makes the system easier to maintain and extend with additional analytical capabilities.

Overall, the Enterprise AI Data Analyst provides an effective solution for assisting users in exploring business data, generating meaningful insights, and supporting business decision-making in an intuitive and interactive manner.

6.2 Future Work

Although the proposed system successfully meets its current objectives, several improvements can be incorporated in future versions to enhance its capabilities.

Future enhancements include:

- Expanding the Transformer model to support a wider range of business questions and analytical intents.
- Integrating Large Language Models (LLMs) to answer more flexible and complex business queries.
- Supporting real-time database connections instead of only CSV and Excel datasets.

- Adding advanced visualizations such as pie charts, treemaps, heatmaps, and geographical maps.
- Implementing predictive analytics and sales forecasting using machine learning techniques.
- Providing personalized dashboards based on different user roles.
- Deploying the application on a cloud platform for multi-user access.
- Improving AI-generated business recommendations using advanced data-driven decision support techniques.
- Enhancing model accuracy by training on larger and more diverse business datasets.

These improvements would increase the system's scalability, flexibility, and practical value for real-world business environments.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, *et al.*, "Attention Is All You Need," in *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS)*, 2017.

[2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[3] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[4] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, "Distributed Representations of Words and Phrases and Their Compositionality," in *Advances in Neural Information Processing Systems*, 2013.

[5] W. McKinney, "Data Structures for Statistical Computing in Python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 56–61.

[6] Python Software Foundation, *Python Documentation*. Available: https://docs.python.org/

[7] PyTorch, *PyTorch Documentation*. Available: https://pytorch.org/docs/

[8] Streamlit, *Streamlit Documentation*. Available: https://docs.streamlit.io/

[9] Plotly Technologies Inc., *Plotly Python Documentation*.
Available: https://plotly.com/python/

[10] Pandas Development Team, *Pandas Documentation*.
Available: https://pandas.pydata.org/docs/

[11] NumPy Developers, *NumPy Documentation*. Available: https://numpy.org/doc/

[12] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed.
Burlington, MA, USA: Morgan Kaufmann, 2011.